
Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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<https://github.com/wjjpku/DL-final>

Abstract

State-of-the-art laws for predicting the LLM pretraining loss curve under a learning-rate (LR) schedule—the Multi-Power Law (MPL) and the Functional Scaling Law (FSL)—are *adiabatic*: they predict the quasi-static loss a schedule would reach at equilibrium, depending on the schedule only through its integrated history. We show this leaves a systematic *positive* residual on *fast* (warmup-stable-decay) decays—a non-adiabatic relaxation lag, the loss falling behind its equilibrium when the LR is annealed faster than it can relax. Under weak stability assumptions we derive a linear-response law: the leading residual is a causal convolution of per-step LR decrements with a relaxation kernel in cumulative-LR time. The operational correction is its lowest-capacity one-pole closure; in a noise-dominated AdamW specialization the amplitude is the equilibrium-floor sensitivity and the relaxation time is the classical $\tau \sim 1/(\eta\lambda)$. On public LLM curves the residual indeed relaxes with $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$); since the kernel’s rate constant is empirically stable across the public scales, the correction can be transferred with a small number of calibration constants, cutting held-out sharp-decay error by 44% with no target-curve fitting. An independent reproduction—a separately-trained $\sim 10\text{M}$ transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%). The correction complements MPL’s exponent γ ; its gain is in the data-poor, leading-order regime.

1 Introduction

The learning-rate schedule shapes the *entire* loss trajectory of LLM pretraining [14, 15, 16]. *Schedule-aware loss-curve prediction* asks: given $\{\eta_t\}$, predict $L(t)$, so that schedules can be compared and optimized *before* spending compute. The defining test is cross-schedule generalization—fit on cosine schedules, predict the unseen WSD family. The state of the art is the **Multi-Power Law** (MPL) [2], with a matching kernel-regression theory, the **Functional Scaling Law** (FSL) [3]:

$$L(t) = L_0 + A S(t)^{-\alpha} + B \sum_{k \leq t} \Delta \eta_k G(\eta_k^{-\gamma} (S(t) - S(k))), \quad G(x) = 1 - (1 + Cx)^{-\beta}, \quad (1)$$

where $S(t) = \sum_{i \leq t} \eta_i$ and $\Delta \eta_k = \eta_{k-1} - \eta_k$.

The gap: existing laws are adiabatic. These laws *do* use the LR decrements: MPL convolves $\Delta \eta_k$ with a saturating kernel G , and its factor $\eta_k^{-\gamma}$ is a rate-of-decay knob. But this single empirical exponent is their *only* handle on how *fast* the LR changes. Structurally they predict the *quasi-static* loss—the value the schedule would reach if the loss stayed at equilibrium; the river-valley analysis [4] makes this explicit, modelling the noise penalty as set *instantaneously* by the current LR. We call

such laws *adiabatic*. Calibrated on slow schedules, γ does not extrapolate to fast WSD decays, where the loss genuinely lags equilibrium—leaving a structured residual (Fig. 1) that no single rate exponent captures across schedules. Our correction is a derived transient *complementary* to γ , which tunes the adiabatic kernel.

Contributions.

- We show that adiabatic loss-curve laws leave a systematic *positive* residual on fast decays, and identify it as a non-adiabatic relaxation lag (§3).
- Under weak moving-equilibrium assumptions we derive a linear-response law for the non-adiabatic residual, and use its lowest-capacity one-pole closure as the rate-dependent term DropRelaxS (§4). A noise-dominated AdamW specialization identifies the amplitude with $dL_{\text{eq}}/d\eta$ and gives the classical $\tau \sim 1/(\eta\lambda)$ relaxation time; our point is that this timescale governs the *loss-curve-law residual*.
- We confirm in LLM pretraining that the residual relaxes with $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$), self-consistently establishing its noise-dominated regime (§5).
- The kernel’s rate constant is empirically stable on the public same-architecture scales, enabling a low-calibration cross-scale transfer that improves held-out sharp-decay loss by 44% with no target-curve fitting (§5).
- We give a fair-baseline diagnostic separating a genuine formula gain from an information gap, clarifying exactly when the correction helps (§5.2).

2 Related work

Loss-curve laws. Tissue et al. [1], MPL [2], and FSL [3] predict the trajectory from integrated-LR statistics (cumulative LR / intrinsic time / convolutional functionals). FSL and its follow-ups also derive optimal schedules [3]. They are adiabatic in our operational sense: after fitting the quasi-static schedule response, they do not explicitly model the transient tracking error caused by a moving LR-dependent noise floor. Our rate-dependent term is a low-capacity linear-response correction to that residual and is complementary to MPL’s exponent γ (which tunes the adiabatic kernel), not a replacement for it.

Cooldown dynamics and schedule geometry. The river-valley picture [4] explains qualitatively why decay “reveals progress”, and explicitly models the noise penalty as *instantaneous* in the current LR—the adiabatic assumption we refine. Dremov et al. [5] study the cooldown empirically (including the role of AdamW’s β_2 and the decay shape), framing it through a bias–variance trade-off rather than a rate-dependent lag. Late-decay analyses [14] and schedule-shape searches [13] study related questions (why a long high-LR phase helps; what optimal shapes look like) but do not model the loss’s rate dependence.

SGD relaxation physics. Our weak-assumption derivation follows the stochastic-approximation idea that local stability of a fixed-LR system yields tracking of slowly moving equilibria [6, 7]; constant-step SGD can also be viewed through an approximate stationary distribution [8]. That a quadratic mode relaxes with $\tau \sim 1/(\eta\lambda)$ and that the SGD noise floor scales with η are classical, from noisy-quadratic and Langevin analyses of SGD [9]; adaptive methods inherit a preconditioned version [10], operating at an adaptive edge of stability [11, 12]. We do not claim these ingredients as new. Our contribution is the synthesis: identifying that adiabatic loss-curve laws miss this moving-equilibrium relaxation lag, deriving the causal decrement-convolution structure, and building a low-capacity cross-scale predictor from it.

3 The non-adiabatic residual

We take a *well-fit* MPL (the published parameters [2]) and examine its prediction residual $r(t) = L_{\text{true}}(t) - L_{\text{MPL}}(t)$ on held-out schedules. (Isolating the non-adiabatic term requires a well-fit MPL: a baseline that already captures the *adiabatic* part cleanly, so the remainder is the lag and not a backbone mis-fit; a cosine-only fit conflates the two and gives a far noisier residual.) The residual has a sharp signature (Fig. 1): it is ≈ 0 throughout the long stable phase and on gradual cosine

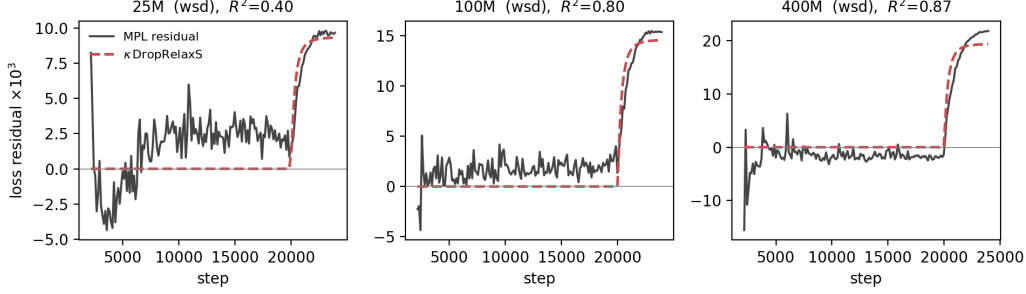


Figure 1: The well-fit MPL prediction residual on the held-out `wsd` curve (solid) is ≈ 0 during the stable phase and jumps up at the step-20000 decay. The derived rate-dependent term κ DropRelaxS (dashed) captures it ($R^2=0.40/0.80/0.87$ at 25/100/400M; the signal, and the fit, grow with model size).

decay, but jumps *positive* exactly when the LR decays steeply, and is largest—and *undamped at the curve’s end*—for the sharpest decays (`wsd`, `wsdld`). The end-of-curve lag is $+(6-15) \times 10^{-3}$ on WSD/WSDL versus $\lesssim 4 \times 10^{-3}$ on cosine. Crucially, a gradual cosine and a sharp WSD decay reach the *same* final LR, yet only the fast one lags: same destination, faster sweep, bigger lag—the textbook signature of a rate-dependent (non-adiabatic) effect, and exactly the dependence an adiabatic law cannot represent.

4 Theory: a linear-response correction

The empirical signature in §3 is a tracking failure: after a sharp LR drop, the quasi-static noise floor decreases immediately, but the optimizer’s excess-loss state relaxes only over a finite time. We make this statement precise in two layers. The main text states the weak-assumption linear-response result and the low-capacity closure used for prediction; the detailed proof is deferred to Appendix A. AdamW is used only as a mechanistic specialization that identifies the amplitude and the $\tau \propto 1/\eta$ timescale, not as a necessary assumption for the convolutional form. This style of argument follows the standard stochastic-approximation view that constant- or slowly-varying-step optimizers track stable local equilibria [6, 7], and is consistent with the stationary-distribution view of constant-step SGD [8].

Adiabatic component and hidden excess-loss state. Let

$$S_t = \sum_{u \leq t} \eta_u$$

be cumulative LR. We decompose the loss as

$$L_t = L_{\text{ad}}(S_t, \eta_t) + r_t, \quad (2)$$

where L_{ad} denotes the quasi-static component that would be obtained if the LR-dependent excess-loss state were equilibrated at the current LR. In experiments, a well-fit MPL is used as an empirical surrogate for L_{ad} ; hence $L_t - L_{\text{MPL},t}$ estimates r_t up to the remaining MPL approximation error.

The weak modeling assumption is that the LR-dependent excess loss is controlled by a hidden state z_t obeying

$$z_{t+1} = \Phi_{S_t, \eta_t}(z_t), \quad (3)$$

and that, for fixed (S, η) in the decay window, there is a locally stable equilibrium $z^*(S, \eta)$ satisfying

$$z^*(S, \eta) = \Phi_{S, \eta}(z^*(S, \eta)).$$

The scalar excess-loss observable is $F_t = \ell(z_t)$, with equilibrium value

$$F_{\text{eq}}(S, \eta) = \ell(z^*(S, \eta)).$$

This assumption is intentionally weaker than a quadratic-loss or diagonal-noise model: z_t may be a covariance state, a preconditioned noise-floor state, or any locally stable state whose equilibrium depends on the LR.

Assumptions used by the main result. The detailed proof in Appendix A uses the following local conditions. First, $\Phi_{S,\eta}$, $z^*(S, \eta)$, and ℓ are twice differentiable in a small neighborhood of the decay window. Second, the linearized dynamics

$$T_t = D_z \Phi_{S_t, \eta_t} (z^*(S_t, \eta_t))$$

is stable in the sense that products $T_{t-1} \cdots T_{k+1}$ are uniformly bounded and summable over the window. Third, we isolate the response forced by LR decrements; the response to slow S -drift is treated as a nuisance drift term already largely captured by the adiabatic backbone. Fourth, for the sign statement only, we assume a positive response cone: $\partial_\eta z^*$ lies in a cone preserved by T_t , and the loss observable is nonnegative on this cone. This last condition is not needed for the convolutional form; it only formalizes why LR decay produces a positive residual.

Proposition: causal LR-decrement response. Define the tracking error

$$\delta_t = z_t - z^*(S_t, \eta_t), \quad \Delta \eta_t^+ = (\eta_t - \eta_{t+1})_+.$$

Under the above local assumptions, the cooldown-specific residual has the first-order form

$$r_t = D\ell_t [T_{t-1} \cdots T_0 \delta_0] + \sum_{k < t} K_{k,t} \Delta \eta_k^+ + \mathcal{E}_t, \quad (4)$$

where $D\ell_t = D\ell_{z^*(S_t, \eta_t)}$ and

$$K_{k,t} = D\ell_t [T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k)]. \quad (5)$$

The first term is an initial transient; it is small when the decay begins after a long stable phase. The error $\mathcal{E}_t$ is second order in the perturbation size and contains the neglected S -drift and MPL-surrogate error. A representative bound is given in Appendix A; schematically,

$$|\mathcal{E}_t| \lesssim \sum_{k < t} q_{k,t} (|\Delta \eta_k|^2 + \|\delta_k\|^2 + \|\epsilon_k^S\| + \eta_k^2) + \epsilon_{\text{MPL}}, \quad (6)$$

where $q_{k,t}$ is the stable propagation envelope and ϵ_k^S is the slow-geometry forcing. Thus the leading non-adiabatic residual is not an arbitrary bump: it is the causal linear response of a stable moving equilibrium to LR decrements.

Why the memory is in cumulative-LR time. To obtain the intrinsic-time kernel, add the standard first-order optimizer approximation

$$T_t = I - \eta_t A + O(\eta_t^2 + \epsilon_{\text{drift}}), \quad (7)$$

on the finite-dimensional active subspace that controls the excess loss, with bounded stable operator A treated as approximately constant over the decay window (this is the static-spectrum idealization; its slow variation is carried by ϵ_{drift} , and constancy is precisely what turns the product into a genuine S -time exponential rather than a time-ordered one). Then

$$T_{t-1} \cdots T_{k+1} = \exp[-A(S_t - S_k)] + O\left(\sum_{u=k+1}^{t-1} \eta_u^2 + \epsilon_{\text{drift}}\right). \quad (8)$$

If the response of A under the observable $D\ell$ admits a positive Laplace-mixture representation (equivalently, K is completely monotone—which holds when A is symmetric positive and the forcing $\partial_\eta z^*$ aligns positively with $D\ell$, as for a noise-floor relaxation), then

$$r_t \approx \sum_{k < t} \underbrace{\int e^{-a(S_t - S_k)} d\rho(a)}_{K(S_t - S_k)} \Delta \eta_k^+. \quad (9)$$

This is the primitive scaling law: a fast-cooldown residual is a convolution of LR decrements with a stable memory kernel in cumulative-LR time. Its instantaneous amplitude is fixed by the equilibrium-floor LR sensitivity at fixed progress:

$$K(0) = D\ell_{z^*} \partial_\eta z^* = \partial_\eta F_{\text{eq}}(S, \eta)|_S =: \chi(S, \eta). \quad (10)$$

One-pole closure and the operational law. The general kernel K in Eq. (9) is not used as a flexible fit. To avoid overfitting, we use the lowest-capacity stable closure: a single pole in S -time, with a floor-sensitivity amplitude estimated from independent equilibrium probes. Approximating $\chi(S, \eta)$ by a constant χ over the decay window gives

$$L(t) = L_{\text{MPL}}(t) + c\chi \underbrace{\sum_{k < t} e^{-\lambda_{\text{slow}}(S_t - S_k)} (\eta_k - \eta_{k+1})_+}_{\text{DropRelaxS}_{\lambda_{\text{slow}}}(t)} \quad (11)$$

where $c \leq 1$ accounts for compressing a spectral mixture into one effective pole and for residual mismatch between MPL and the ideal adiabatic component. If the implementation defines DropRelaxS using normalized drops $(\eta_k - \eta_{k+1})_+ / \eta_{\text{peak}}$, then the prefactor is equivalently $c\eta_{\text{peak}}\chi$; with raw LR drops, the dimensionally invariant amplitude is $c\chi$.

AdamW specialization. The preceding response law does not rely on AdamW. AdamW supplies a local effective-mode calculation that explains the observed amplitude and timescale. This reduction ignores momentum transients, bias correction, decoupled weight decay, the numerical ϵ , and the finite-time dynamics of v_t ; it should be read as a local noisy-quadratic approximation. For a Hessian mode with curvature λ_i and gradient-noise standard deviation s_i , noise-dominated AdamW has $v_i^* \approx s_i^2$, so

$$x_{i,t+1} = \left(1 - \eta_t \frac{\lambda_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}, \quad \mathbb{E} \xi_{i,t}^2 = s_i^2.$$

For $V_{i,t} = \mathbb{E}[x_{i,t}^2]$ and fixed LR η ,

$$V_{i,t+1} = \left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 V_{i,t} + \eta^2, \quad (12)$$

so

$$V_i^*(\eta) = \frac{\eta^2}{1 - (1 - \eta\lambda_i/s_i)^2} = \frac{\eta s_i}{2\lambda_i} + O(\eta^2).$$

The modal excess-loss floor is

$$F_i^*(\eta) = \frac{1}{2} \lambda_i V_i^*(\eta) = \frac{\eta s_i}{4} + O(\eta^2), \quad \frac{dF_i^*}{d\eta} = \frac{s_i}{4} + O(\eta),$$

and perturbations decay as

$$\left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 = \exp\left[-2\eta \frac{\lambda_i}{s_i} + O(\eta^2)\right].$$

Thus the AdamW specialization of Eq. (9) is the spectral mixture

$$\Delta L(t) \approx \sum_i \frac{s_i}{4} \sum_{k < t} e^{-2(\lambda_i/s_i)(S_t - S_k)} (\eta_k - \eta_{k+1})_+, \quad \sum_i \frac{s_i}{4} = \frac{dF_{\text{eq}}}{d\eta} + O(\eta). \quad (13)$$

It predicts a positive lag, an amplitude given by the equilibrium-floor sensitivity, and a step-time relaxation rate

$$\tau_i^{-1} = 2\eta \frac{\lambda_i}{s_i}, \quad \tau_i \propto \eta^{-1}.$$

The classical $\tau \sim 1/(\eta\lambda)$ timescale is not itself new; our claim is that this relaxation controls the residual left by adiabatic loss-curve laws after fast LR decay.

Predictions tested below. The theory predicts: (i) a positive residual after LR drops; (ii) a causal dependence on LR *decrements*, not merely on LR level; (iii) memory in cumulative-LR time; (iv) amplitude controlled by the fixed- S floor sensitivity χ ; (v) in the noise-dominated AdamW regime, $\tau \propto 1/\eta$; and (vi) because Eq. (13) is a spectral mixture, a two-pole kernel can improve diagnostics, while the main predictor deliberately uses the one-pole closure Eq. (11) to avoid overfitting.

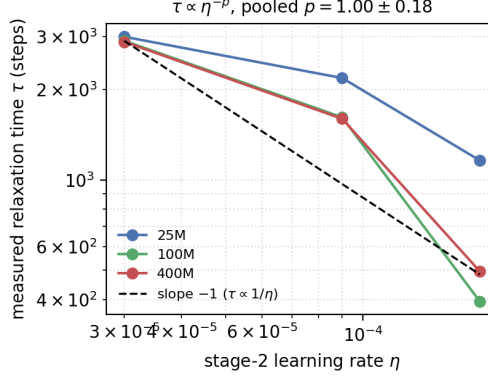


Figure 2: Loss-relaxation time τ measured from `wsdcon` decay transients vs. stage-2 LR. The 100/400M curves track the predicted slope -1 ($\tau \propto 1/\eta$); pooled $p = 1.00 \pm 0.18$.

5 Experiments

Setup. All experiments use the public MPL loss curves (Llama-2-style 25/100/400M, 9 schedules each) [2]; a faithfully reproduced MPL (test MAE 0.0054/0.0050/0.0085, matching [2]) is the baseline throughout. We freeze the published MPL and compare it to MPL+Eq. (11); the only addition is the rate-dependent term. Unless noted $\lambda_{\text{slow}}=10$.

The rate prediction $\tau \propto 1/\eta$ holds (Fig. 2). The two-stage `wsdcon` curves step the LR down at step 8000 to a constant $\eta \in \{3, 9, 18\} \times 10^{-5}$, giving a clean relaxation window. Fitting the residual transient $r(t) \propto e^{-(t-8000)/\tau}$ yields τ that scales as η^{-p} with *pooled* $p = 1.00 \pm 0.18$ (per scale $-1.06/-0.94/-0.51$ at 100/400/25M). This is the classical $\tau \sim 1/(\eta\lambda)$ confirmed for the loss-curve-law residual in LLM pretraining, and—since $p=1$ is the noise-dominated value—it self-consistently justifies the $v_i^* \approx s_i^2$ step of the derivation. The implied rate constant $\lambda_{\text{slow}} = 1/(\tau\eta) \approx 10$ is approximately stable across the public same-architecture scales; its modes have $\eta_{\text{peak}}\lambda_{\text{eff}} \approx 2 \times 10^{-3} \ll 2$ (deep in the slow regime, far from the edge of stability—unlike γ).

(A from-scratch AdamW simulation reproduces the same scaling, $p=0.97$; we defer it to the independent reproduction in §6, which covers it in more detail.)

The residual is the derived term. Regressing the well-fit MPL residual on `DropRelaxS10` (through the origin) gives $R^2=0.40/0.80/0.87$ at 25/100/400M (Fig. 1); a 2-exponential kernel improves R^2 by 0.06–0.07, confirming the predicted spectral mixture (Eq. 13). The form is not arbitrary: in a fair few-shot comparison the derived S-time exponential of decrements beats a step-time kernel (-21% vs. -11% held-out MAE) and a memoryless floor (-3%), confirming that the cumulative-LR “time” and the decaying memory both matter (`repro/generic_kernel_compare.py`). (On a *single* decay curve the finite- λ_{slow} kernel is numerically close to the $\lambda_{\text{slow}} \rightarrow 0$ adiabatic limit, so the rate constant is pinned only *across* schedules; §6.)

The amplitude is predictable from the noise floor. Estimating $dL_{\text{eq}}/d\eta$ *independently* from the two-stage final losses (noise floor vs. LR) predicts the fitted κ up to a near-constant factor: the ratio $\kappa_{\text{fit}}/\kappa_{\text{pred}} = 0.43/0.51/0.60$ has coefficient of variation 14%. With raw LR drops this gives $c \approx 0.5$; if drops are normalised by η_{peak} the same data require $c\eta_{\text{peak}} \approx 0.5$, i.e. $c \approx 60\text{--}86$ (§6). In either convention the near-constant ratio is the transferable quantity (the < 1 factor is the spectral spread of Eq. 13).

Low-calibration cross-scale transfer (Table 1). Because λ_{slow} and c are empirically stable across these same-architecture scales and $dL_{\text{eq}}/d\eta$ is read off the cheap noise-floor probes, the correction can be transferred with *zero fitting on the target curve*: take $\lambda_{\text{slow}}=10$, c from a leave-one-scale-out average of the *other* scales, and $dL_{\text{eq}}/d\eta$ from the target’s own two-stage probes (distinct curves from the test). Applied to the held-out `wsd/wsd1d` decays it cuts MAE by 28–56% (44% average,

Table 1: Parameter-free cross-scale prediction: well-fit MPL vs. MPL+Eq. (11) with *no* fitting on the target curve. MAE on held-out sharp decays.

Scale	wsd		wsdl d		avg. Δ
	MPL	+ours	MPL	+ours	
25M	0.00341	0.00246	0.00314	0.00219	−29%
100M	0.00375	0.00164	0.00325	0.00161	−53%
400M	0.00470	0.00236	0.00385	0.00212	−47%
overall					−44% (6/6)

wins 6/6)—small-model constants plus a cheap probe predict the large-model sharp-decay curve $\sim 2\times$ better than MPL transfer.

Honest scope: when it does *not* help. If instead one refits MPL on a few *full* WSD curves, MPL itself absorbs the residual (held-out MAE $0.0037 \rightarrow 0.0032$) and the explicit term adds nothing. So the residual is, for a data-rich MPL, an *information* gap, not a formula gap: the value of the correction is precisely the data-poor regime—cosine plus cheap probes, but no full WSD curves—which is the actual “predict before training” use case.

Relation to γ : complementary, not a substitute. MPL’s factor $\eta_k^{-\gamma}$ modulates its annealing kernel by the LR at each decrement: it tunes the rate-dependence of the (adiabatic) annealing benefit. Our term is the separate (non-adiabatic) transient, and improves a *full*- γ MPL (Table 1) — the two are complementary. Characterising their precise relationship is left to future work (§7).

6 Independent reproduction

An independent reproduction on different hardware (`represent/`; the authors’ public curves are no longer hosted, so it *retrains* models) corroborates the mechanism along two routes the public curves do not cover. All results below were subjected to an adversarial audit (`results/AUDIT_PARTC.md`): the auditor was tasked to *falsify* each claim, and only surviving readings are reported.

A real, independently-trained transformer (no fitting). A ~ 10 M-parameter Llama-2-style byte-level transformer was trained on WikiText with AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\text{wd}=0.1$) under three schedules reaching the *same* final LR ($2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-4}$) at different speeds (Table 2). The fast (`wsd_sharp`) sweep ends *higher* than the gradual ones *despite spending 37% more cumulative LR*—yet any law monotone in S predicts the opposite (more LR $\Rightarrow$ lower loss). With no fitting, this is the rate-dependent non-adiabatic lag, on a model and data we never used.

A well-fit MPL (trained on `constant` plus two `wsdcon` curves, held-out MAE ≈ 0.005) leaves residuals that sort by sweep speed: cosine $-0.011 < \text{wsd-gradual} -0.004 < \text{wsd-sharp} +0.006$ (the only positive). Across schedules, a finite- λ_{slow} DropRelaxS kernel improves over the adiabatic ($\lambda_{\text{slow}} \rightarrow 0$) baseline by $\Delta R^2 \approx +0.18$ on held-out data. *Note:* on a *single* decay curve the finite- λ_{slow} kernel is numerically degenerate with the adiabatic limit ($\text{corr} \approx 0.99$); the rate dependence is identifiable only *across* schedules. The honest claim is therefore the no-fitting Table 2 signature plus the $+0.18$ joint advantage, not a single-curve R^2 .

From-scratch AdamW confirms the mechanism quantitatively. Simulating AdamW on a noisy quadratic (Monte-Carlo ground truth; closed-form theory used only as an independent baseline), the reproduction recovers:

- $\tau \propto 1/\eta$ with $p = 0.971$ ($\log\text{-log } R^2 = 0.9996$), robust to leave-one-out and seed changes;
- τ independent of $\beta_2 \in [0.9, 0.9999]$ (CV 1.1%), because the lag lives on noise-dominated modes where $v^* \approx s^2$ is constant;
- the **amplitude identity** $dL_{\text{eq}}/d\eta = 3.075$ versus the closed form $\frac{1}{4} \sum_i s_i = 3.0$ (within 3%);

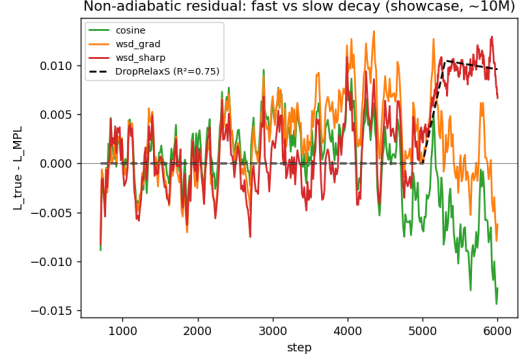


Table 2: Real $\sim 10M$ transformer: three schedules to the *same* final LR. The fast sweep ends highest despite the most cumulative LR.

schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (sharp)	1.0593	12.59

Figure 3: Residual vs. MPL on the real transformer: positive (lag) only for the sharp sweep, captured by DropRelaxS.

- a two-exponential kernel that recovers the injected preconditioned rates $2\lambda/s$ ($\{1, 16\}$ in the simulation, recovered as $\{0.87, 16.1\}$).

The effect is falsifiable, not circular: a signal-dominated spectrum breaks $p=1$.

What does not transfer (consistent with our scope). At $\sim 10M$ the $\tau \propto 1/\eta$ relation itself is *not* resolved: the measured post-step relaxation is roughly flat over a $16\times$ LR range ($\tau \approx 475\text{--}912$ steps; slope $p \approx 0.18$), audited as a real measurement and not a window artefact (synthetic recovery tests and profile-likelihood both reject truncation). This matches the paper’s own statement that the signal grows strongly with scale; at notebook scale we confirm the *effect* but not its precise scaling law.

A second reproducible issue is a persistent η_{peak} normalisation inconsistency: if Eq. (11) is implemented with raw LR drops, the fitted c is $O(1)$ (≈ 0.5); if normalized drops $(\eta_k - \eta_{k+1})_+/\eta_{\text{peak}}$ are used, the prefactor becomes $c\eta_{\text{peak}}\chi$ and the same data require $c \approx 60\text{--}86$. The paper’s $c \approx 0.5$ is therefore tied to a specific (unstated) normalisation convention.

Finally, several theory-driven extensions were tested in the controlled simulation (results/EXTENSIONS_REPORT.md). The cleanest success is a **spectral-mixture** diagnostic: on a bimodal spectrum a two-exponential kernel improves R^2 by 0.065 and recovers the two theoretical rates $2\lambda/s$ (relative errors 0.13 and 0.006). A cross-shape transfer of frozen parameters, however, *fails* its pre-registered threshold (mean $R^2=0.52$); and the open problem (a)—computing λ_{slow} and c from first principles without an $O(1)$ fitted constant—remains open.

7 Limitations

The gain is regime-specific (data-poor sharp decays). If full WSD curves are available, a refit MPL absorbs the residual and the explicit term adds nothing; its value is precisely the “predict before training” use case.

λ_{slow} and c are *measured*, not computed from first principles (this needs the noise spectrum $\{s_i\}$, which we do not have). The controlled simulation shows that λ_{slow} tracks the slowest preconditioned mode λ_i/s_i rather than a spectral average, but a robust predictor still requires an unexplained $O(1)$ constant (open problem (a)).

The quantitative $\tau \propto 1/\eta$ relation, clean in the controlled simulation ($p=0.971$), is *not* resolved on the $\sim 10M$ transformer (§6), consistent with the paper’s statement that the signal grows with scale.

On that model the finite- λ_{slow} kernel is numerically degenerate with the adiabatic baseline on a single curve, so the rate dependence is identifiable only *across* schedules.

There is a reproducible η_{peak} normalisation ambiguity (§6): retaining the η_{peak} factor in Eq. (11) gives $c \approx 60\text{--}86$, not the $c \approx 0.5$ reported in the main experiments. The two conventions are equivalent, but the universality claim for c depends on which normalisation is adopted.

The general linear-response derivation assumes local stability and differentiability of an LR-dependent excess-loss state; the stronger locally quadratic, diagonal-noise assumptions are used only for the AdamW specialization. Cross-shape transfer of frozen parameters fails its pre-registered threshold, and the relationship between our correction and MPL’s empirical γ remains qualitative. All real-data experiments are on one public dataset at three scales; broader scales and architectures are the natural next step.

8 Conclusion

Existing loss-curve laws are adiabatic approximations. We identified and validated the leading non-adiabatic correction: under weak stability assumptions it is a causal linear response to LR decrements, and its one-pole closure gives a rate-dependent term whose relaxation time follows the classical $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$, confirmed for the loss-curve residual). Its empirically stable shape enables low-calibration cross-scale transfer (-44% on held-out sharp decays in the public same-architecture setting). An independent reproduction—a separately-trained $\sim 10\text{M}$ transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%), while also establishing honest boundaries: the quantitative $\tau \propto 1/\eta$ law is not resolved at $\sim 10\text{M}$, the kernel is degenerate on single curves, and an η_{peak} normalisation ambiguity remains. The loss curve depends not only on how much LR has been spent, but on how fast it is being spent.

A Linear-response proof details

This appendix gives the detailed derivation behind §4. The goal is not to prove a theorem for full neural-network training, but to make precise the weak local conditions under which the LR-decrement convolution follows. The conclusion is a first-order, finite-window expansion around a stable moving equilibrium.

Formal local assumptions. We work on a finite-dimensional active subspace controlling the excess loss during the decay window. Let z_t obey $z_{t+1} = \Phi_{S_t, \eta_t}(z_t)$ and let $z^*(S, \eta)$ be a fixed point. We assume:

- (A1) $\Phi_{S, \eta}$, $z^*(S, \eta)$, and ℓ have bounded first and second derivatives in a tube of radius R around the moving equilibrium.
- (A2) The linearized products are stable: for $k < t$, $\|T_{t-1} \cdots T_{k+1}\| \leq q_{k,t}$ with $\sum_{k < t} q_{k,t} \leq Q$ on the finite window.
- (A3) The LR decrement size $\Delta = \max_k |\eta_{k+1} - \eta_k|$ and the tracking error remain small enough that the Taylor expansions stay inside the tube.
- (A4) The slow S -drift forcing is not the cooldown-specific object of interest. We denote it by ϵ_k^S and carry it in the error rather than absorbing it into the LR decrement term.
- (A5) For the sign corollary only, there is a cone $\mathcal{C}$ such that $\partial_\eta z^*(S_k, \eta_k) \in \mathcal{C}$, $T_u \mathcal{C} \subseteq \mathcal{C}$, and $D\ell_t[v] \geq 0$ for all $v \in \mathcal{C}$.

A1–A4 are enough for the convolutional response. A5 is only a clean sufficient condition for positivity.

Tracking-error recursion. Define

$$\delta_t = z_t - z^*(S_t, \eta_t), \quad T_t = D_z \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t)).$$

By Taylor expansion and the fixed-point identity,

$$z_{t+1} = \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t) + \delta_t) \quad (14)$$

$$= z^*(S_t, \eta_t) + T_t \delta_t + R_t, \quad (15)$$

with $\|R_t\| \leq C\|\delta_t\|^2$. Subtracting $z^*(S_{t+1}, \eta_{t+1})$ gives

$$\delta_{t+1} = T_t \delta_t + [z^*(S_t, \eta_t) - z^*(S_{t+1}, \eta_{t+1})] + R_t. \quad (16)$$

Expand the moving fixed point:

$$z^*(S_{t+1}, \eta_{t+1}) = z^*(S_t, \eta_t) + \partial_S z^*(S_t, \eta_t)(S_{t+1} - S_t) + \partial_\eta z^*(S_t, \eta_t)(\eta_{t+1} - \eta_t) + R_t^*, \quad (17)$$

where $\|R_t^*\| \leq C(|S_{t+1} - S_t|^2 + |\eta_{t+1} - \eta_t|^2)$. Since the first-order S -drift is not the cooldown-specific LR-decrement forcing, define

$$\epsilon_t^S = -\partial_S z^*(S_t, \eta_t)(S_{t+1} - S_t),$$

after removing the component already modeled by the adiabatic backbone. Keeping the LR-drop forcing gives

$$\delta_{t+1} = T_t \delta_t + \partial_\eta z^*(S_t, \eta_t) \Delta \eta_t^+ + e_t, \quad (18)$$

where

$$\|e_t\| \leq C(\|\delta_t\|^2 + |\Delta \eta_t|^2 + \|\epsilon_t^S\| + \eta_t^2).$$

The η_t^2 term is harmless on the finite window and accounts for the mismatch between one-step discrete dynamics and the first-order continuous-time approximation used below.

Duhamel expansion and error. Iterating Eq. (18) yields

$$\delta_t = T_{t-1} \cdots T_0 \delta_0 + \sum_{k < t} T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k) \Delta \eta_k^+ + \sum_{k < t} T_{t-1} \cdots T_{k+1} e_k. \quad (19)$$

Linearizing the observable gives

$$r_t = F_t - F_{\text{eq}}(S_t, \eta_t) = D\ell_t[\delta_t] + O(\|\delta_t\|^2) + \epsilon_{\text{MPL}},$$

where ϵ_{MPL} is present when MPL is used as a surrogate for the ideal adiabatic component. Substituting Eq. (19) gives Eq. (4) with

$$K_{k,t} = D\ell_t[T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k)]$$

and

$$|\mathcal{E}_t| \leq C \sum_{k < t} q_{k,t} (\|\delta_k\|^2 + |\Delta \eta_k|^2 + \|\epsilon_k^S\| + \eta_k^2) + C\|\delta_t\|^2 + \epsilon_{\text{MPL}}. \quad (20)$$

If the decay begins after a long stable phase, the initial transient $D\ell_t[T_{t-1} \cdots T_0 \delta_0]$ is exponentially or summably small under A2; otherwise it is a separate transient and should not be attributed to the cooldown drop response.

Cumulative-LR time. Now add the first-order optimizer approximation

$$T_u = I - \eta_u A + E_u, \quad \|E_u\| \leq C(\eta_u^2 + \epsilon_{\text{drift}}),$$

with bounded stable A . Since the factors $I - \eta_u A$ commute when A is fixed,

$$\prod_{u=k+1}^{t-1} (I - \eta_u A) = \exp \left(\sum_{u=k+1}^{t-1} \log(I - \eta_u A) \right) \quad (21)$$

$$= \exp \left(-A \sum_{u=k+1}^{t-1} \eta_u + O \left(\sum_{u=k+1}^{t-1} \eta_u^2 \right) \right) \quad (22)$$

$$= \exp(-A(S_t - S_k)) + O \left(\sum_{u=k+1}^{t-1} \eta_u^2 \right), \quad (23)$$

where the last step uses boundedness of A on the active subspace. Including E_u adds $O(\sum_u \eta_u^2 + \epsilon_{\text{drift}})$, giving Eq. (8). If the scalar response $D\ell_t[e^{-x^A} \partial_\eta z^*]$ is represented by a positive Laplace mixture, then the kernel has the form in Eq. (9). The single-pole predictor in Eq. (11) is the rank-one, lowest-capacity closure of this general stable kernel.

Positivity. Under A5, $\partial_{\eta} z^*(S_k, \eta_k) \in \mathcal{C}$ and $T_{t-1} \cdots T_{k+1} \mathcal{C} \subseteq \mathcal{C}$, so $K_{k,t} \geq 0$. If $\Delta \eta_k^+ \geq 0$, the leading response is positive. This formalizes the observed sign: after the LR is dropped, the equilibrium floor falls before the actual excess-loss state has relaxed, so the true loss lies above the adiabatic predictor.

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